

HORNSBY GIRLS HIGH SCHOOL



YEAR 11 MATHEMATICS

2009

Preliminary Assessment 1

Time Allowed: 60 minutes

Instructions:

- Attempt all questions
- Start a new page for each question
- The marks for each question are indicated
- Show all necessary working
- Use blue or black pen.
- Marks may be deducted for untidy or badly arranged work
- Board approved calculators may be used

Outcomes Assessed:

- Demonstrates confidence in using mathematics to obtain realistic solutions to problems (P1)
- Provides reasoning to support conclusions which are appropriate to the context (P2)
- Performs routine arithmetic and algebraic manipulation involving surd and simple rational expressions (P3)
- Chooses and applies appropriate arithmetic and algebraic techniques (P4)
- Understands the concept of a function and the relationship between a function and its graph (P5)

Outcomes	Questions			Totals
Basic Algebra and Arithmetic	Q1:	Q2:	Q4:	/34
Real Functions of a Real Variable	Q3:	Q5:		/21
	Total			/55

This assessment task constitutes 20% of the final Preliminary Course Assessment

Question 1: (12 marks)

START A NEW PAGE

Marks

- (a) Evaluate, correct to 2 significant figures

$$(\sqrt{5}-1)^2 \times \sqrt{\frac{1}{3 \cdot 2} + \frac{3}{7}}$$

2

- (b) Express $0.43\bar{8}$ as a common fraction.

2

- (c) Factorise $250x^3 + 16y^6$.

2

- (d) Show that $\frac{1}{3+\sqrt{3}} + \frac{1}{3-\sqrt{3}}$ is a rational number.

2

- (e) Simplify $\frac{-4a^3}{3b^7} \times \frac{a^2}{8b^2} \times \frac{-6b^4}{a^8}$.

2

- (f) Solve $|2x-3|=1$.

2

Question 2: (11 marks)

START A NEW PAGE

- (a) Solve for x : $5x-3 < 3x+3$

1

- (b) Solve for x : $\frac{x-5}{3} - \frac{x+1}{4} = 5$.

3

- (c) (i) Make y the subject of $\frac{y-x}{z} = \frac{y+z}{w}$.

2

- (ii) Hence or otherwise, find the value of y when $z=2$, $w=3$ and $x=1$.

2

- (d) Solve simultaneously:

$$3x-4y=-3$$

$$3x+y=12$$

2

- (e) Find the value of a if $x^2+5x=(x+a)^2-\frac{25}{4}$.

1

Question 3: (10 marks)

START A NEW PAGE

Marks

(a) For the function $f(x) = x\sqrt{x}$

(i) State the domain of $f(x)$.

1

(ii) Find $f(4)$.

1

(iii) Find an expression for $f(a)$ and $f(ab)$.

2

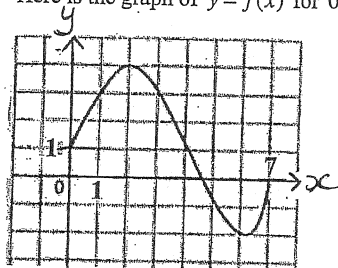
(iv) Hence, show that $f(a)f(b) - f(ab) = 0$.

2

(b) If $f(x) = 2x - x^2$, find $f(x+1) - f(1)$ in its simplest form.

2

(c) Here is the graph of $y = f(x)$ for $0 \leq x \leq 7$.



(i) Evaluate $f(2)$.

1

(ii) Find the range of $y = f(x)$.

1

Question 4: (11 marks)

START A NEW PAGE

Marks

(a) Simplify $\frac{c^2 - 9}{(c - 3)^2} + \frac{9c + 9}{(c - 3)(c + 1)}$.

2

(b) (i) Show that $(x - 1)^3 - (x + 1)^3 = -2(3x^2 + 1)$.

3

(ii) Hence solve $(x - 1)^3 - (x + 1)^3 = -56$.

2

(c) Solve $|x + 3| = 2x - 1$.

2

(d) Solve for x : $x^2 - a^2 \geq 0$ for $a \geq 0$.

2

Question 5: (11 marks)

START A NEW PAGE

(a) Show that the function $f(x) = \frac{-x}{x^2 - 4}$ is an odd function.

2

(b) Find the domain and range of $y = \frac{1}{x - 3}$.

2

(c) Consider the functions $h(x) = \sqrt{1 - x}$ and $g(x) = \sqrt{1 + x}$.

(i) Find the domain of $h(x)$ and $g(x)$

2

(ii) Hence, or otherwise, find the domain of $f(x) = \sqrt{1 - x} + \sqrt{1 + x}$.

1

(iii) Determine whether $y = f(x)$ is even, odd or neither.

2

(d) Consider the function defined by $f(x) = \begin{cases} x + 2 & x < -1 \\ x^2 & |x| \leq 1 \\ -x + 2 & x > 1 \end{cases}$.

Find the value of $f(-3) + f(3) - f\left(\frac{1}{2}\right)$.

2

END OF PAPER

Year 11 Mathematics Preliminary Assessment 1 2009 - Solutions

Question 1

(a)

$$(\sqrt{5}-1)^2 \times \sqrt{\frac{1}{3.2} + \frac{3}{7}} = 1.315269...$$

$$= 1.3(2sf)$$

(b)

$$\text{Let } x = 0.\dot{4}3\dot{8}$$

$$x = 0.438438438...$$

$$1000x = 438.438438...$$

$$999x = 438$$

$$x = \frac{438}{999}$$

$$x = \frac{146}{333}$$

(c)

$$250x^3 + 16y^6 = 2(125x^3 + 8y^6)$$

$$= 2[(5x)^3 + (2y^2)^3]$$

$$= 2(5x + 2y^2)(25x^2 - 10xy^2 + 4y^4)$$

(d)

$$\frac{1}{3+\sqrt{3}} + \frac{1}{3-\sqrt{3}} = \frac{3-\sqrt{3}+3+\sqrt{3}}{(3+\sqrt{3})(3-\sqrt{3})}$$

$$= \frac{6}{3^2-3}$$

$$= \frac{6}{6}$$

$$= 1$$

(e)

$$\frac{-4a^3}{3b^7} \times \frac{a^2}{8b^2} \times \frac{-6b^4}{a^8} = \frac{24a^5b^4}{24a^8b^9}$$

$$= a^{-3}b^{-5}$$

$$= \frac{1}{a^3b^5}$$

(f)

$$|2x-3|=1$$

$$2x-3=1 \text{ or } 2x-3=-1$$

$$2x=4 \text{ or } 2x=2$$

$$x=2 \text{ or } x=1$$

Question 2

(a)

$$5x-3 < 3x+3$$

$$2x < 6$$

$$x < 3$$

(b)

$$\frac{x-5}{3} - \frac{x+1}{4} = 5$$

$$\frac{4(x-5) - 3(x+1)}{12} = 5$$

$$4x-20-3x-3=60$$

$$x-23=60$$

$$x=83$$

(c) (i)

$$\frac{y-x}{z} = \frac{y+z}{w}$$

$$w(y-x) = z(y+z)$$

$$wy - wx = zy + z^2$$

$$wy - zy = wx + z^2$$

$$y(w-z) = wx + z^2$$

$$y = \frac{wx + z^2}{w-z}$$

(c) (ii)

$$y = \frac{wx + z^2}{w-z}$$

$$y = \frac{3(1) + 2^2}{3-2}$$

$$y = 3 + 4$$

$$y = 7$$

(d)

$$3x - 4y = -3 \dots (1)$$

$$3x + y = 12 \dots (2)$$

$$(2) - (1):$$

$$y + 4y = 12 + 3$$

$$5y = 15$$

$$y = 3$$

Sub into (1)

$$3x - 4(3) = -3$$

$$3x - 12 = -3$$

$$3x = 9$$

$$x = 3$$

$$\therefore x = 3, y = 3$$

(e)

$$x^2 + 5x = \left(x + \frac{5}{2}\right)^2 - \frac{25}{4}$$

$$a = \frac{5}{2}$$

Question 3

(a) (i)

$$\text{domain} : x \geq 0$$

(a) (ii)

$$\begin{aligned} f(4) &= 4\sqrt{4} \\ &= 8 \end{aligned}$$

(a) (iii)

$$f(a) = a\sqrt{a}$$

$$f(ab) = ab\sqrt{ab}$$

(a) (iv)

$$\begin{aligned} f(a)f(b) - f(ab) &= a\sqrt{a} \times b\sqrt{b} - ab\sqrt{ab} \\ &= ab\sqrt{ab} - ab\sqrt{ab} \\ &= 0 \end{aligned}$$

(b)

$$\begin{aligned} f(x+1) - f(1) &= [2(x+1) - (x+1)^2] - [2(1) - (1)^2] \\ &= 2x + 2 - x^2 - 2x - 1 - 2 + 1 \\ &= -x^2 \end{aligned}$$

(c) (i)

$$f(2) = 4$$

(c) (ii)

Range: $-2 \leq y \leq 4$

Question 4

(a)

$$\begin{aligned}\frac{c^2 - 9}{(c - 3)^2} + \frac{9c + 9}{(c - 3)(c + 1)} &= \frac{(c - 3)(c + 3)}{(c - 3)^2} + \frac{9(c + 1)}{(c - 3)(c + 1)} \\ &= \frac{c + 3}{c - 3} + \frac{9}{c - 3} \\ &= \frac{c + 12}{c - 3}\end{aligned}$$

(b) (i)

$$\begin{aligned}(x - 1)^3 - (x + 1)^3 &= x^3 - 3x^2 + 3x - 1 - (x^3 + 3x^2 + 3x + 1) \\ &= -6x^2 - 2 \\ &= -2(3x^2 + 1)\end{aligned}$$

ALTERNATIVE METHOD

$$\begin{aligned}(x - 1)3 - (x + 1)^3 &= [(x - 1) - (x + 1)][(x - 1)^2 + (x - 1)(x + 1) + (x + 1)^2] \\ &= (-2)(x^2 - 2x + 1 + (x^2 - 1) + x^2 + 2x + 1) \\ &= (-2)(x^2 + x^2 + x^2 - 2x + 2x + 1 - 1 + 1) \\ &= -2(3x^2 + 1)\end{aligned}$$

(b) (ii)

$$\begin{aligned}(x - 1)^3 - (x + 1)^3 &= -56 \\ -2(3x^2 + 1) &= -56 \\ 3x^2 + 1 &= 28 \\ 3x^2 &= 27 \\ x^2 &= 9 \\ x &= 3 \text{ or } -3\end{aligned}$$

(c)

$$\begin{aligned}|x + 3| &= 2x - 1 \\ x + 3 &= 2x - 1 & x + 3 &= -2x + 1 \\ x &= 4 & 3x &= -2 \\ \text{Test } x &= 4 & x &= \frac{-2}{3} \\ LHS &= 7 & \text{Test } x &= \frac{-2}{3} \\ RHS &= 1 & LHS &= \frac{4}{3}, \quad RHS = \frac{-1}{3} \\ \therefore x &= 4 & \therefore x &= \frac{-2}{3} \text{ is NOT a solution}\end{aligned}$$

(d)

$$\begin{aligned}x^2 - a^2 &\geq 0 \\ (x - a)(x + a) &\geq 0\end{aligned}$$

Method 1:

Case 1: $(x - a) \geq 0 \cap (x + a) \geq 0$ [both positive or zero]

$$x \geq a \cap x \geq -a$$

$$\therefore x \geq a$$

Case 2: $(x - a) \leq 0 \cap (x + a) \leq 0$ [both negative or zero]

$$x \leq a \cap x \leq -a$$

$$\therefore x \leq -a$$

Solution is $x \leq -a$ or $x \geq a$

Method 2:

From graph, $x \geq a$ or $x \leq -a$

Question 5

(a)

$$f(x) = \frac{-x}{x^2 - 4}$$

$$f(-x) = \frac{-(-x)}{(-x)^2 - 4}$$

$$= \frac{x}{x^2 - 4}$$

$$= -f(x)$$

$$\therefore f(-x) = -f(x)$$

$\therefore f(x)$ is odd function

(b)

Domain : $x \neq -3$

Range : $y \neq 0$

(c) (i)

$$h(x) = \sqrt{1-x}$$

Domain : $x \leq 1$

$$g(x) = \sqrt{x+1}$$

Domain : $x \geq -1$

(c) (ii)

$$f(x) = \sqrt{1-x} + \sqrt{1+x}$$

Domain :

$$x \geq -1 \text{ AND } x \leq 1$$

$$\therefore -1 \leq x \leq 1$$

(c) (iii)

Let $y = f(x)$

$$f(x) = \sqrt{1-x} + \sqrt{1+x}$$

$$f(-x) = \sqrt{1-(-x)} + \sqrt{1-x}$$

$$= \sqrt{1+x} + \sqrt{1-x}$$

$$= \sqrt{1-x} + \sqrt{1+x}$$

$$= f(x)$$

$\therefore y = f(x)$ is an even function

(d)

$$f(-3) = -3 + 2$$

$$= -1$$

$$f(3) = -1$$

$$f\left(\frac{1}{2}\right) = \frac{1}{4}$$

$$\therefore f(-3) + f(3) - f\left(\frac{1}{2}\right) = -2\frac{1}{4}$$